

Innovation Insights Quarterly: Q1 2026

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The most durable innovations do more than solve an immediate problem. They set the stage for future breakthroughs. From a novel method for rare-earth extraction to more accurate dead cell identification to AI infrastructure in space the innovations from this quarter show how present-day limits in materials, biology, and compute capacity are directly shaping the next generation of solutions. While their paths to impact differ, each demonstrates how near-term problem solving can expand what becomes possible over time.

Replacement mitochondria restore damaged cells

Researchers have identified a method to restore the energy production of old or damaged cells via mitochondrial replacement.¹ Mitochondria are the structures that generate usable energy for cells. The study paired flower-shaped nanoparticles with stem cells, prompting the stem cells to produce twice their normal number of mitochondria. When placed nearby, the stem cells transferred the excess mitochondria to the weakened cells, replenishing cellular energy capacity.

Why it matters: Mitochondrial decline is linked to aging, heart disease and neurodegenerative disorders. Population studies show that people over 70 carry nearly 60% more mitochondrial DNA mutations on average than those under 40.² While still early, this work points to a potential path to improve cell function across the body, improving the quality of late life.

Robotics and AI reshape fulfillment operations

Large e-commerce companies are increasingly deploying automation tools to get packages to customers as fast as possible and at lower cost. Some advancements include robotic arms that sort packages, AI systems that help managers identify and avoid bottlenecks, and augmented reality glasses that assist drivers with navigation and package identification. Together, these examples illustrate the growing role of automation in large-scale logistics.

Why it matters: Logistics is increasingly becoming a proving ground for robotics and AI. At Amazon, the largest logistics operator in the world, these technologies now assist roughly three-quarters of deliveries and are translating directly into faster throughput. The company's Shreveport, LA, facility, which contains roughly 10 times as many robots as a typical warehouse, can process packages 25% faster.³ These improvements offer a glimpse of how AI and robotics-led productivity gains could compound into sustained GDP expansion.

For illustrative purposes only.

Satellites power space-based AI infrastructure Space-based computing is being explored as a response to AI's rising energy needs⁴

Orbital satellites power space-based AI infrastructure

Companies are now exploring constructing data centers in space as a way to address AI's rising energy needs. While designs aren't finalized, one possible option involves deploying solar-powered satellites equipped with AI chips and connecting them via optical links, forming an AI computing network in orbit.

Why it matters: As AI models grow more energy-intensive, power has become a key constraint. In space, solar panels can be eight times more productive than they are on Earth, producing power nearly continuously and reducing the need for batteries and terrestrial grids. This new location for data centers will build on a satellite infrastructure that is already extensive and improving, with large scale satellite constellations with one cluster already counting more than 9,000 satellites in orbit.⁵

Levitation enables cell identification

Scientists have developed an electromagnetic device that can sort and identify cells by levitating them to different

heights.⁶The device separates cells based on physical properties such as density, allowing researchers to distinguish between healthy and cancerous cells without exposing samples to harsh chemicals or centrifugal force.

Why it matters:Many biomedical techniques depend on isolating healthy, viable cells from mixed samples. Less invasive sorting methods can improve the reliability of diagnostics and drug testing by reducing cell damage. In testing, researchers were able to raise sample viability from 50% live cells to more than 90%, producing cleaner inputs for downstream analysis.

Special viruses extract rare earth elements

Researchers have developed a low-impact method for mining rare earth elements (REEs) using engineered viruses.⁷The approach uses bacteriophages (viruses that infect bacteria but are harmless to humans and ecosystems) modified with specialized proteins that allow the virus to collect and extract REEs from mine drainage water with minimal waste.

Why it matters:Rare earth processing is concentrated in only a few countries because existing extraction methods generate up to 2,000 metric tons of waste per ton produced,⁸poisoning waterways and degrading soil. The subsequent chemical processing also produces significant air pollution. Because this method is both low-cost and environmentally friendly, it could be deployed by a wider range of countries with rare earth reserves, reducing environmental damage while lowering reliance on a small number of global suppliers.

The company examples discussed below are included for illustrative purposes, selected based on the scale, visibility, and maturity of their publicly disclosed initiatives. They are intended to illustrate broader technological and economic trends, rather than to represent a comprehensive survey of participants or an assessment of relative investment attractiveness

Endnotes

Source: Texas A&M University Engineering. "Recharging the powerhouse of the cell." November 20, 2025.

Source: BMC Genomics. Independent impacts of aging on mitochondrial DNA quantity and quality in humans. November 21, 2017.

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Source: New Atlas. "Biomining viruses deliver rare earth elements but no toxic horrors of mining." December 1, 2025.

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